

Inferring dividend yield and discount rates from option prices

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This note details how one can extract a dividend yield and a discount curve from a surface of option prices written on an underlying security S_t . Let $T_1 < T_2 < \dots < T_m$ denote the maturities of the various options traded. For each maturity we also observe option prices for a set of strikes. For each available strike and maturity, we observe both a put price, $P(T_i, K_j^i)$ and a call option price, $C(T_i, K_j^i)$. To back out the dividend yield q and the spot rates $r_{T_i}, i = 1, \dots, m$, we should choose a set of pairs of calls and puts for each maturity and with strikes around At-The-Money (ATM). Let n denote the number of option pairs we choose on each side of ATM (e.g. $n = 5$). We can back out q and r_{T_i} for all $i = 1, \dots, m$ by minimizing the following objective function over q and $r_{T_i}, i = 1, \dots, m$

$$\sum_{i=1}^m \sum_{j=1}^{2n} \left(P(T_i, K_j^i) - P^{PC}(C(T_i, K_j^i), q, r_{T_i}) \right)^2, \quad (1)$$

where

$$P^{PC}(C(T_i, K_j^i), q, r_{T_i}) = C(T_i, K_j^i) + K_j^i e^{-r_{T_i} T_i} - S_0 e^{-q T_i}, \quad (2)$$

is the put price obtained from the call option market price $C(T_i, K_j^i)$ using the put-call parity with the dividend and the spot rate for maturity T_i given by q and r_{T_i} , respectively.

Filtering of the data:

It is also recommended to apply a filter on the data:

- Remove options prices with bid price equal to 0.
- Remove ITM options (can be debated but makes good sense if the underlying security is a stock index like the SPX).
- Various arbitrage restrictions like in Carr and Madan (2005), Cousot (2007), Kokholm (2016).

References:

- Carr, P. and Madan, D. B. (2005). A note on sufficient conditions for no arbitrage. *Finance Research Letters*, 2(3):125–130.
- Cousot, L. (2007). Conditions on option prices for absence of arbitrage and exact calibration. *Journal of Banking and Finance*, 31(11):3377–3397.
- Kokholm, T., Pricing and hedging of derivatives in contagious markets. *Journal of Banking and Finance*, 2016, **66**, 19–34.